

Q1. You have a dataset X of 10 binary observations. Your Null Hypothesis is that the Bernoulli parameter is 0.5, while the alternate hypothesis is that this parameter is below 0.5. Your test-statistic $T(X)$ is the fraction of '1's, and your rejection region is defined by $(T(X)-0.5 < \epsilon)$. What will be possible values of ϵ , if you are to reject the null hypothesis at 5% level of significance?

Q2. In the above case, suppose you have obtained 6 heads. What will be the p-value if i) the alternate hypothesis is that parameter is below 0.5, ii) alternate hypothesis is that parameter is different from 0.5?

Q3. Let $(X_1, \dots, X_9)$ are random samples from $N(\mu, 4)$. To test $H_0: \mu=2$ vs $\mu>2$, find the p-value when the observed sample mean is 2.3.

Q4. You are given 10 observations from a distribution which is supposed to be $\text{Gamma}(10, 5)$. It is known that the second parameter is correct, but someone claims that the first one is greater than 10. What is the smallest level of significance at which you can agree with the claim? [1.95 2.7 1.99 1.51 3.79 2.44 2.54 1.19 2.25 1.81]

Q5. You have two datasets: $A=[2.3 \ 3.1 \ 2.8]$ and $B=[3.0 \ 3.6 \ 5.2]$. The null hypothesis is that, they follow the same distribution. Test this hypothesis using the Permutation test over any 10 (randomly chosen) permutations, at 5% level of significance.

Q6. You have a sequence of observations [3.4, -1.5, 7.2, -2.8, -0.4, 6.5]. Would you like to model this with a single high-variance Gaussian, or a mixture of two low-variance Gaussians? Explain your answer using AIC and BIC.